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SURVEY ARTICLE

A Review of Augmented Reality Heads Up Display in Vehicles: Effectiveness, Application, and Safety

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ABSTRACT
This paper reviews the evolution of driver displays, focusing on the transition to Augmented Reality Heads-Up Display (AR HUD) in vehicles. It compares AR HUD with traditional HUD, emphasising safety, cognitive workload, and user experience. Particular attention is given to AR HUD's role in automated vehicles, especially during driver handover from automated to manual driving. Findings suggest AR HUD can enhance safety and user experience when applied effectively but may increase distraction and cognitive load if misused. Benefits include improved navigation through reduced cognitive load, decreased inattentional blindness, and better obstacle detection, such as identifying pedestrians. A notable gap in the literature is that AR HUD is often studied as an extension of the main driver display rather than as a primary display. This review highlights AR HUD's potential as the sole display, emphasising the need for further research. Overall, AR HUD presents transformative possibilities for enhancing vehicle user experience.

KEYWORDS
Augmented reality; Heads up display; AR HUD; ADAS; automated vehicles; driver display

1. Introduction

Driving plays a substantial part of life for the majority of households. In fact, over 78% of households owned at least one car in 2021 (National Travel Survey, 2021). With such a high percentage of ownership, driving is an everyday task for many. During driving, traditionally, the driver would have some sort of information cluster, instrument panel or dials in front of them. As described by Sun et al. these dials typically show the speed using a speedometer and an engine tachometer (Sun et al., 2020). These could be complemented with another dial for the fuel level and water temperature.

The dials slowly transitioned into part digital, where the current speed was displayed with a type of digital output. For instance, the 1983 Renault 11 had a fully digital display, but even before that the 1976 Aston Martin Lagonda was the first (Braithwaite-Smith, 2023). The advancement of displays throughout the years enabled more and more vehicles to have digital displays.

Then, there is the heads up display (HUD). It has evolved from the reflector sight which was a technology that originated from the 1900s (Jarrett, 2017). Afterwards, the technology has moved into aircraft cockpits, and later into vehicles, with the first one being commercially available was the 1988 Oldsmobile Cutlass Supreme (Johnson, 2019). But what is a heads-up display or HUD? It is a type of display and an addition to dials or digital display. It is another way of displaying various driving information's such as speed of the vehicle. It uses optics to project information onto the windscreen in front of the driver's field of view and it enables the drivers gaze to remain at the outside world. There is also the option during the build process to set the distance of the HUD image, meaning it can appear that it is a few metres ahead of the driver, which can reduce the reaccommodating time between the driver looking at the HUD display and the outside world compared to having the HUD display closer to the driver (Ward & Parkes, 1994). Commercial HUDS do have their issues. They do not superimpose information between the real and virtual world. Because there can be a mismatch between these virtual projections and the real scene, it can create issues in terms of cognitive workload (Park & Kim, 2013). This is where Augmented Reality Heads-up Display or AR HUD comes in.

Augmented Reality Heads Up Display (AR HUD) is an evolution of a traditional HUD, where the real environment is supplemented with virtual objects or information using computer technology (Husár et al., 2022; Park & Kim, 2013), which is similar to HUD, however most of the information displayed, were already part of the dashboard, such as the speed. AR HUD is able to provide a larger amount of information (Park et al., 2013). Because AR HUD can interact with real objects, it can for example alert drivers of an obstacle, highlight road signs or upcoming turns and lane information. The purpose of this literature review is to explore the current and future use of AR HUD.

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With this technology, vehicle manufacturers are able to implement solutions where the AR HUD augments traffic signs many metres away, highlight upcoming hazards such as cars braking and aid in navigation by directly projecting arrows to specific turns. A description by Deng et al. states that AR HUD is a system that allows the user to see "navigation and warning information directly through the windshield" (Deng et al., 2021). Various authors, highlight the fact that traditional HUD can only produce images at a fixed distance, which can cause frequent refocusing as opposed to AR HUD which can appear at a variable distance (Deng et al., 2021; Schneider, 2021; Skirnewskaja & Wilkinson, 2022; Wang et al., 2022). AR HUD can have the ability to increase drivers' situational awareness since their gaze is kept on the road rather than looking down. AR HUD could even display non driving critical content like entertainment, especially when looking at more automated vehicles (Riegler et al., 2021).

A literature review focused on the visualisation of AR HUD was conducted in 2022; in this the author explored the user experience and trust. This was review from a perspective of specific features like highlighting routes and object highlights (Kettle & Lee, 2022). This literature review expands on that and considers motion sickness, more highlights and various other angles for trust and on top of that covers other aspects like the size of the AR HUD area for effectiveness. In fact, the review states that more should be researched around inclusive design and visual impairment which is what this literature review covered around disabilities. In addition, this literature review explores research works into inattentional blindness and presents a different literature gap.

The scope of this paper is around specific use cases where AR HUD's performance and benefit can be measured and where AR HUD can be compared to traditional HUD. The effectiveness and performance section includes navigation for a specific use case and goes onto expand some key concepts that AR HUD can potentially help with. The other focus is with automated vehicles with two big focus trust and handovers. The former can potentially be a big hurdle to get over which makes it an important topic from the AR HUD perspective and the latter will be an increasingly common action as more and more semi- automated vehicles release where control must be handed back to the user. This is a new phenomenon for driving and one that is important to explore from the AR HUD perspective to make as safe as possible.

To guide the review and frame the discussion, clear research questions have been formulated. These questions are intended to address the key areas of interest and provide a structured approach to understanding the impact and potential of AR HUD systems in vehicles. The primary research questions are:

- How does AR HUD compare to traditional HUD in terms of effectiveness, performance, and user experience?
- What impact does AR HUD have on cognitive load, inattentional blindness, and obstacle detection?
- How can AR HUD be effectively integrated into automated vehicles, particularly during handover scenarios?
- What additional considerations, such as psychological impacts, visibility, and usability, should be accounted for when implementing AR HUD?

While this is a review paper and not an empirical study, these questions help in structuring the review and focusing on key areas of interest.

The remainder of this paper is structured as follows: Section 2 details the literature review strategy. Section 3 explores the application of AR HUD in navigation, including SAE levels and handover applications where AR HUD can assist in guiding user focus. Section 4 examines psychological aspects such as inattentional blindness, trust, anxiety, and colour blindness. Section 5 discusses design and features, including obstacle highlighting, visibility, 3D aspects, size, and controls, all critical for AR HUD performance. Section 6 addresses unaddressed challenges and future trends, particularly the potential for AR HUD to serve as the main and only display in vehicles, eliminating traditional driver displays.

2. Literature review strategy

To ensure a comprehensive and systematic review of the literature on AR HUD in vehicles, a rigorous search and selection strategy has been employed. The following outlines the methodology and criteria for including or excluding studies from this review.

2.1. Search strategy

The literature search was conducted using several academic databases, including Google Scholar, IEEE Xplore, and ScienceDirect. The search aimed to cover a broad range of relevant studies, ensuring comprehensive coverage of the topic. The following keywords and phrases were used to search for relevant literature:

- "Augmented Reality Heads-Up Display" (AR HUD)
- "AR HUD effectiveness and performance"
- "AR HUD navigation"
- "AR HUD inattentional blindness"
- "AR HUD cognitive load"
- "AR HUD obstacle detection"
- "Autonomous vehicles AR HUD handover"
- "SAE levels and AR HUD"
- "Trust in AR HUD systems"
- "Psychological aspects of AR HUD"
- "AR HUD visibility and 3D"
- "AR HUD size and controls"
- "AR HUD and colour blindness"
- "AR HUD vs HUD"

2.2. Inclusion and exclusion criteria

The inclusion criteria for selecting studies were:

- Relevance: The study must be directly related to AR HUD in vehicles.

- Publication Date: Preference was given to studies published in the last 10 years to ensure contemporary relevance.
- Peer-Reviewed: Only peer-reviewed articles, conference papers, and a few websites from specific manufacturer and company.

The exclusion criteria were:

- Irrelevance: Studies not directly related to the specified AR HUD topics were excluded.
- Non-peer-reviewed Sources: Articles such as opinion pieces, non-peer-reviewed papers, and non-scientific articles were not considered.
- Duplicate Studies: Multiple studies reporting the same findings were filtered to avoid redundancy.

2.3. Analysis and synthesis

Each selected study was analysed to extract relevant data regarding the effectiveness, performance, cognitive load, navigation, inattentional blindness, obstacle detection, and other pertinent aspects of AR HUD. For studies on autonomous vehicles, focus was placed on SAE levels, trust, and handover scenarios. Additional considerations such as psychological impacts, visibility, 3D aspects, size and controls, and disability issues (e.g., colour blindness) were also reviewed.

The extracted data was then synthesised to identify patterns, draw comparisons, and highlight key findings and gaps in the literature. This comprehensive approach ensured that the review encompassed a broad and relevant spectrum of the current state of AR HUD research in vehicles.

3. Application

3.1. Navigation

There are several sources that compare the effectiveness of HUD compared to AR HUD. Bolton et al. has focused research with navigation elements and found that compared to traditional HUD distance-to-turn navigation performed worse than AR landmark box presentation. The results of this research concluded that the response times and success rates were improved by 43.1% and 26.2% respectively, in favour of the AR HUD solution. In addition to this the drivers stated a "significant reduction of workload" (Bolton et al., 2015). All of this shows that there are serious benefits in using AR HUD technologies for navigation. It is worth noting that in this research landmark boxes were essentially AR HUD boxes that superimposed over specific landmarks, however as part of the research AR arrows showing turns were also used to compare against traditional HUD. The research concluded that landmark boxes highlighted specific landmarks at required intersections, interestingly worked even better than AR HUD arrows (Bolton et al., 2015), as shown in Figure 1.

Further study can be found by Wu et al, in their study around landmark based car navigation that uses a full-windshield HUD. The study they conducted is similar to the one by Bolton et al. as they also highlighted specific landmarks for instance street signs to aid in navigation, but it was approached from a technical standpoint around distortion correction. The study concluded that highlighting these landmarks is possible, but the system needs to consider distortion related to the car windshield, location of sensors and the angle. In the research they argued that in traditional navigation systems the cognitive load increases, since the user has to map the information between the phone/navigation system and the real world. That is where AR HUD can help (Schneider, 2021; Wang et al., 2022; Wu et al., 2009). This study did not go into psychological effects, however later research and studies in this paper do explore more around cognitive load and trust.

To expand Bark et al. had published specific research around AR HUD and navigation. In general, the research results were positive and found that users were able to pinpoint turn locations earlier with AR HUD systems, along with keeping eyes up longer. They have expanded that the design of the system could be improved by taking advantage of environmental landmarks (Bark et al., 2014), which is similar to the research what Bolton et al. suggested and Wu et al. highlighted around the usage of landmark boxes (Bolton et al., 2015; Wu et al., 2009) Other suggestion mentioned that the "preview" stage should also be visible to ensure the user knows what is coming up after the current manoeuvre (Chauvin et al., 2023).

Research, that explored effective HUD and AR HUD design suggested that AR graphics might not be better than other HUD interfaces. It compared fixed and conformal arrows for navigation and found that distraction, driving impact, navigation, trust and viewing all scored worse than normal HUD. We must bear in mind that the participants were asked subjectively rather than measured questions such as "Using this interface had a positive impact on my driving" (Gabbard et al., 2019). However, actual measurements showed that participants would glance/look at AR HUD more often and for longer than HUD. This in conclusion means that HUD resulted in lower workload and better usability. Simply projecting AR HUD graphics are "not necessarily an inherently better user experience, and spatially locating directional graphics into the forward roadway can cause more workload in some instances" (Gabbard et al., 2019); conformal and screen fixed examples are shown in Figure 2. The study suggest that much more work should be done to compare and test the benefits of AR HUD compared to HUD. Especially further examination of visual attention in different road conditions and unexpected events should be explored (Gabbard et al., 2019).

3.2. SAE levels

Fully automated vehicles are on the rise (Autonomous Vehicles Market to Surpass USD 2324.5 Billion by 2030 on Account of Rising Consumer Demand and Increasing R&D Investments | Research by SNS Insider, 2023). Whilst current commercial automated vehicles are very limited and typically SAE level 2 (user supervises and fallback ready)

Figure 1. Experimental conditions showing HUD images superimposed on road scene (from left to right in order: conventional, AR arrows, landmark arrows and landmark boxes (Bolton et al., 2015).

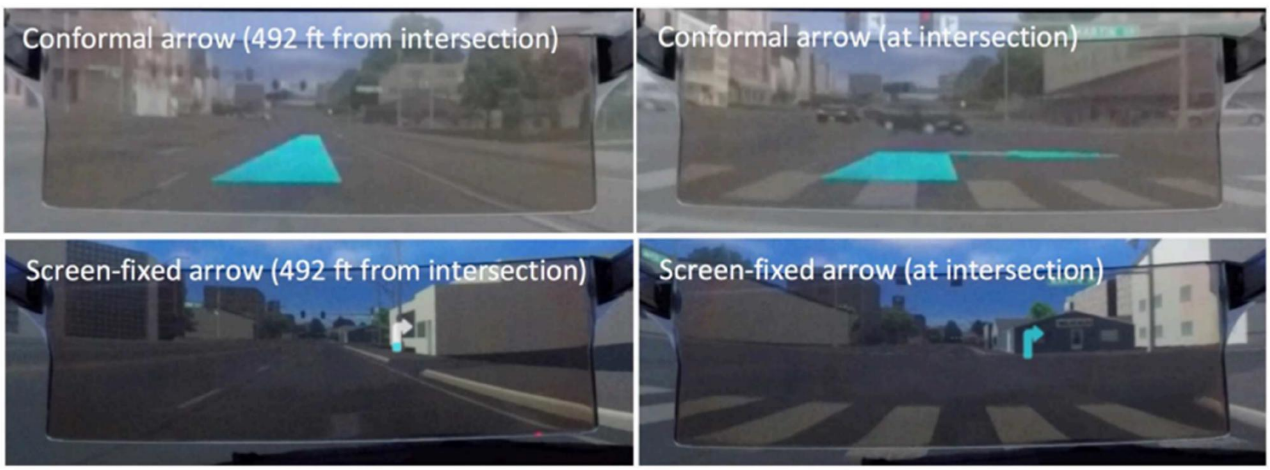

Figure 2. Showing solution with AR HUD arrow (Gabbard et al., 2019).

(International, S, SAE J 3016 - Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, 2023). A very limited number of systems are currently on track to achieve SAE Level 3 (fallback ready) (International, S, SAE J 3016 - Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, 2023); Table 1 summarises the information from SAE International J3016 regarding the different levels of automation.

Mercedes is one the first to bring out an SAE Level 3 driver assist system, specifically in US and more specifically in Nevada for their Mercedes-Benz EQS and S class vehicles soon (Tarantola, 2023). There are exceptions, because there are specific companies such as Waymo using Jaguar I-Pace vehicles operating in specific cities in US at an SAE Level 4. This means that a user can order a driverless taxi and once it arrives, get into the back seat with no-one at the front (Waymo, 2023).

3.3. Handover

In SAE Level 3 which is the one car manufacturers are aiming for, as discussed above (Tarantola, 2023), handover is a key part. Described in SAE J3016, Level 2 requires the user to supervise, and handover (fallback-ready) can happen at any time. Level 3 should be fall-back-ready where handover might need to happen (International, S, SAE J 3016 - Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, 2023). In a study a comparison was made between traditional driving and using AR HUD. The experiment included a longitudinal and lateral malfunction took place. Essentially using AR HUD resulted in fewer crashes by having a significantly lower takeover time where the user had to take control. In addition, AR HUD increases trust and usability rating and no increase in workload was found (Feierle et al., 2022). This cognitive load was highlighted as both negative and positive previously in this paper, however when it is positive and conforms to human computer interaction it could improve cognitive load (Bauerfeind et al., 2021; Bram-Larbi et al., 2021; Cheng et al., 2023; Ma et al., 2021; Schneider, 2021; Teng et al., 2023). The research into handovers and AR HUD, concluded that handovers/take-overs, which are an inherent parts of SAE Level 2 and Level 3 driving, are faster and safer when using AR HUD compared to traditional methods such as driver display (Feierle et al., 2022;

Table 1. Levels of automations (International, S, SAE J 3016 - Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, 2023).

SAE level		Lateral and longitudinal vehicle motion control	Object and event detection and response (OEDR)	Fallback	Operation design domain (area of operation)
0	No automation	Driver	Driver	Driver at all times	Not applicable
1	Driver assistance	Driver and system	Driver	Driver	Limited
2	Partial automation	System	Driver	Driver	Limited
3	Conditional automation	System	System	Fallback-ready (user to become driver during fallback)	Limited
4	High automation	System	System	System	Limited
5	Full automation	System	System	System	Unlimited

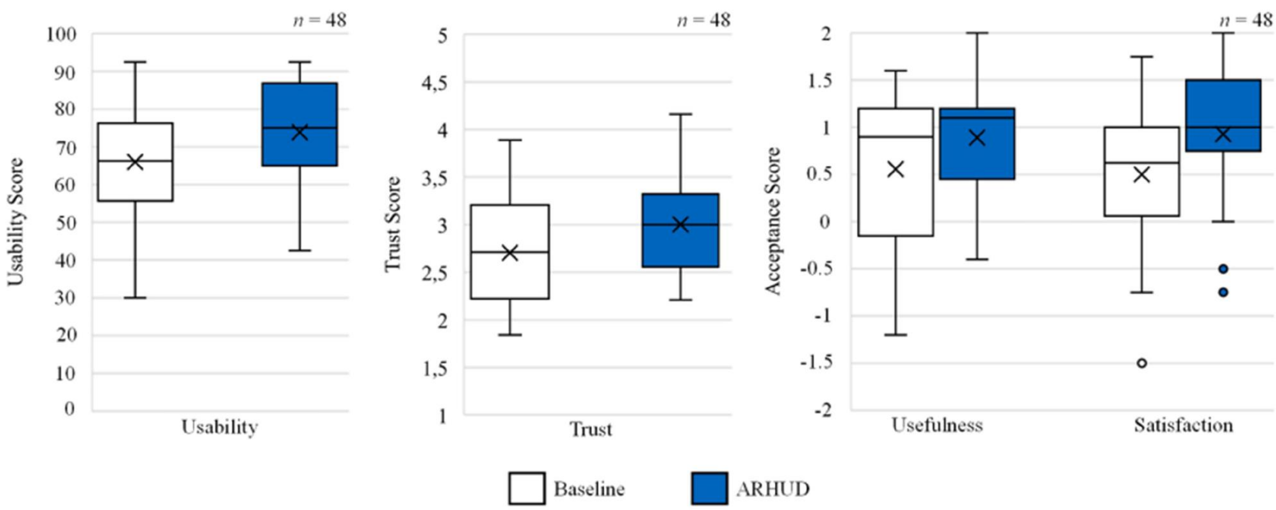

Figure 3. Usability score, trust, usefulness and satisfaction scores (Feierle et al., 2022).

Jing et al., 2022; Wu et al., 2024). This means that AR HUD in automation capable vehicles can promote safety, but on top of that Figure 3 shows that other metrics measured and tested for all performed, on average, better with AR HUD (Feierle et al., 2022).

Related to this as a counterpoint, there was a study around AR HUD and impact on drivers' response to unexpected events. In this study surprise events took place that participants were not aware of. In this situation several positions of HUD-low, mid, high and HDD (head down display) were tested. All scenarios were unexpected and there were more crashes than no crashes recorded. The study was fairly limited in terms of sample size and somewhat inconsistent across these tests, however it seems that AR HUD did not reduce the number of crashes in this study, but it did conclude that the display location may impact the outcome (Smith et al., 2020). It is worth noting that this study was manual driving specifically, so the results where automated driving is directly tested with AR can vary. Other studies highlighted that the way it is implemented needs to be carefully considered to ensure improved handovers (Detjen et al., 2022). More studies and research need to focus on testing, cognitive loads and crash avoidance to see how much AR HUD helps with these as well as the position of the AR HUD system.

Specifically focusing on automated driving and handovers a conference proceedings document concluded an experiment with transition from automated to manual mode with HUD and with AR HUD. Drivers' reactions were typically sharper after a handover request (referred to as takeover here) when compared to manual driving. This includes brake pedal speed and increased vehicle dynamics (the value of acceleration in particular). Fundamentally drivers preferred to have greater distance to vehicles ahead after a takeover. An interesting fact is that straight after takeover, most drivers take less than one second to turn the steering wheel by a significant angle, probably to check that they are indeed in control.

In this study AR had no influence on the takeover itself, however after takeover the results show that driver smooth out the control of the car better and especially if traffic is slowing. The control of the brake pedal and resulting acceleration is smoother. Which can be attributed to the augmented highlight of the vehicle in front with AR HUD. Therefore, the study concludes that AR HUD "has a positive influence on the operation level of driving" (Langlois & Soualmi, 2016). Related to this is that after the takeover the driver has an increased risk of missing an exit or a lane to take, explained by more limited situation awareness after takeover. This seems to be alleviated somewhat with AR HUD, where the driver is more aware of the lane change that must be made by having direct information augmented/projected. In conclusion in this study, AR HUD does not have an effect on the action of takeover, but after it has taken place, AR HUD helps with lane change manoeuvres, which are more anticipated and the control of the vehicle,

Table 2. Takeover/handover times (Langlois & Soualmi, 2016).

Event	Without AR HUD	With AR HUD	*P* value
Duration between takeover request and hands on wheel	M = 3.12s SD = 1.24s	M = 3.13s SD = 1.42s	0.645
Duration between takeover request and pressing a button	M = 4.63s SD = 1.87s	M = 5.09s SD = 2.58s	0.24
Duration between hands on the wheel and pressing a button	M = 1.49s SD = 1.63s	M = 1.44s SD = 1.36s	0.262
Duration between takeover request and the first action on a pedal	M = 7.55s SD = 4.47s	M = 8.95s SD = 7.83s	0.301
Duration between hands on wheel and the first action of the accelerator pedal	M = 4.63s SD = 3.18s	M = 5.12s SD = 3.25s	0.872
Duration between hands on wheel and the first action of the brake pedal	M = 14.01s SD = 10.65s	M = 11.68s SD = 9.25s	0.576

which is smoother (Langlois & Soualmi, 2016). A summary of some of these results can be found in Table 2, showing the median time and standard deviation for both without and with AR HUD.

On the same topic, Gerber et al. (2023) in a qualitative subject matter interview with experts, discuss the duration of non-driving related activities (NDRA) and how long these should and could be in an automated vehicle. Experts agree that reminders should be given to keep the user engaged but did not agree on a specific frequency with some recommending every minute or two, whilst others stating that it should be situation-based. The paper explores how motoric states like body posture, sensory states like distractions and cognitive states like the ability to task switch. The paper presents concerns and argues that maintaining alertness was crucial for fallback/takeover. There are various factors that fall within this that vehicle original equipment manufacturers (OEMs) should do to mitigate risks like limiting physical interactions or designing the vehicle infotainment with NDRA in mind.

4. Psychological

4.1. Inattentional blindness

The next research investigates inattentional blindness. But what is it? Initially explored by Neisser in 1975, by giving participants two videos on top of each other, where they had to count the number of ball passes. Then during the game, the male players were slowly replaced by female players and around half the participants did not notice any changes (Neisser & Becklen, 1975). This highlights the "looking without seeing" or "perceiving vs attending" (Mack, 2003).

A comparison was done by Wang et al. where they explored inattentional blindness then and compared pedestrian visibility with and without AR HUD highlights. It found that "AR HUD reduces inattentional blindness when pedestrians are augmented but encourages inattentional blindness when pedestrians are not augmented." (Wang et al., 2022) It goes into looking at high and low workloads and found that high workload through AR HUD induces inattentional blindness (Chen et al., 2023; Smith et al., 2021; Wang et al., 2022), which will need to be considered in future implementations with countermeasures. It covers 3 types of augmented conditions listed below with the results shown in Table 3. The table shows that if AR HUD is used to highlight everything but pedestrians, it causes the most inattentional blindness, but if in addition it highlights pedestrians then it is the best performer in reducing inattentional blindness.

Table 3. Showing inattentional blindness incidents with and without AR HUD (Wang et al., 2022).

Augmented condition	Incidence of inattentional blindness (%)
No AR-HUD	47.75
AR HUD-NP	63.25
AR HUD-P	32.48

- No AR HUD,
- AR HUD-NP for highlighting with bounding boxes traffic lights, road signs and vehicles
- AR HUD-P for highlighting all of the above in addition to pedestrians

This research also used AR bounding boxes that can direct drivers' attention to the pedestrians or in the previous example landmarks (Bark et al., 2014; Schneider, 2021; Wang et al., 2022; Wu et al., 2009). One gap it raises is that no research has examined the effect of cognitive loads on the occurrence of inattentional blindness in AR HUD driving tasks (Wang et al., 2022).

4.2. Cognitive load

A study around driving performance and cognitive load found that "AR-HUD with interfaces that conform to human-computer interaction principles and visual design rules could improve cognitive resource allocation and promote driving safety" (Ma et al., 2021). The research went into measuring gazes, blink frequency, steering amongst other values comparing it between no AR HUD and two different versions of AR HUD. It found that sweeping, blinking and vehicle motion measurements could be enhancing cognitive resource efficiency with the use of AR HUD. However, "an overly monolithic and centralised design would hinder drivers' driving performance" (Ma et al., 2021). This means that the study could not fully answer the question around AR HUD and driving performance. It went onto explain the importance of the previously mentions human-computer interaction principles and how AR HUD must be very carefully designed in order to facilitate early detection of hazards and improve drivers' performance. In order to achieve this the study suggests that this research should help promote a standardised AR HUD interface (Ma et al., 2021). Other sources highlighted that AR HUD can help reduce cognitive load when designed correctly (Bauerfeind et al., 2021; Bram-Larbi et al., 2021; Cheng et al., 2023; Schneider, 2021; Teng et al., 2023). Although, some others have highlighted the opposite, when the system designed incorrectly either with type of graphics used

(Schneider, 2021) or quantity of information (Feierle et al., 2019). Clearly then, more user trials are required for a definitive answer regarding cognitive load, one which has been highlighted as a difficult subject to measure (de Oliveira Faria, 2020). Especially when some uses could include AR projected advertisements (Wang et al., 2024), further increasing the amount of information that could be displayed. This workload could be measured and contrasted against traditional HUD with the Detection Response Task method which combines reaction times and misses (Maag et al., 2023).

4.3. Trust

Von Sawitzky et al. (2019) in their study found that system transparency is one of the main ways to increase trust. This includes elements such as indication on the AR HUD with an arrow of an upcoming turn, which was explained as a twofold benefit. The indication reassures participants that automated driving is still active, but also allows them to prepare for turns physically which can aid with reducing potential motion sickness. They explain that previewing upcoming navigations is more advantageous than real-time indication which might not be processed fast enough. This point goes back to cognitive load and shows that perhaps it is better to give user upcoming information on AR HUD rather than real time guidance. The study found that in order to reach a high level of trust only world-fixed items are needed to be displayed on AR HUD (for example, the route that a vehicle should take like the next turn) since the information is overlayed on fixed positions of the road or explanatory information (Colley et al., 2023; Koo et al., 2015). Other items such as map or a preview map can be in other displays such as digital driver display or infotainment, as they do not directly correlate with real-world objects or structures. Additional point raised was regarding seating position, as future automated vehicles might shift drivers seating position, in which case further study should be completed on how that shift affects trust and their field of view. Abdi et al. (2015) proposed an Augmented Reality traffic information system. The paper mainly focused on the technical aspects of implementation and computational costs; however, it did cover some elements around users. The paper advises AR HUD to be used to show the status of driver assistance systems focusing on the direct field of view. The proposal highlights that this method increases safety and builds trust with the users. Although with these types of projections, there is a chance that AR HUD incorrectly labels or misses elements. In those cases, the paper did not find it to impair drivers' responses even if they were missed. Other sources claim that visualising all views including windshield and side windows, as a user might not be familiar with all the sensors within the car, and this could increase trust (Colley et al., 2022). On this point there is always some level of uncertainty with labels. Research into uncertainty highlights that it might be advantageous to be upfront with the user and represent the detected visual elements with some level of uncertainty display, but also that designers should be careful in implementation, especially in critical situations (Jung et al., 2015). Although real-time uncertainty is difficult, using deep learning applications the industry can advance and make use of high-speed quantification methods to enable uncertainty displays (Wang et al., 2024).

4.4. Anxiety

From a psychological perspective and effects around using AR HUD system, Hwang et al, conducted a study around these effects and found that with AR HUD, individuals with anti-personal anxiety might benefit from AR HUD systems where the system increases relaxation. The study had its limitations such as limited number of participants and using a simulation where the keyboard was used for breaking instead of an actual simulation with pedals, therefore more research is suggested not only around visual safety and effectiveness, but also the psychological factors AR HUD systems might help with (Hwang et al., 2016). Issues around anxiety could be coupled with models proposed which detects driver emotions (Li et al., 2022) and some actions that can be done to address them (Liu et al., 2023).

4.5. Colour blindness

An extremely important topic to expand upon is disability. Visual impairment is the main one that could affect this. In terms of driving and AR HUD, no research suggested a specific condition that would stop an individual from seeing a HUD, whilst still legally be able to drive. This is true for situations where the user would still have to complete some of the driving tasks like SAE Level 2 or SAE Level 3 (International, S, SAE J 3016 - Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, 2023). However, for even more automated applications, the visibility and a wider range of disabilities will need to be researched to ensure AR HUD elements can be applied in, as an inclusive way as possible. For the sake of conciseness then, this paper only highlights colour blindness which is the main issue that AR HUD application needs to tackle in context of still being able to legally drive. This is a serious consideration as according to Colour Blind Awareness, in the UK alone there are approximately 3 million colour blind people and worldwide that translates to around 1 in 12 men and 1 in 200 women affected and in total that number is around 300 million (About Colour Blindness, 2022). One proposal around this, is that AR HUD graphics could be highly customisable so that they can have specific colour-blind friendly colours if the user has colour blindness. Another suggestion was an overlay for traffic lights for instance that helps enhance visibility to those with colour blindness (He et al., 2022). We can take other research around this and apply it to an automotive setting. Tanuwidjaja et al. (2014), proposed a wearable augmented reality device for colour blindness specifically. The paper explores certain issues people with colour blindness face, such as putting on clothes that match

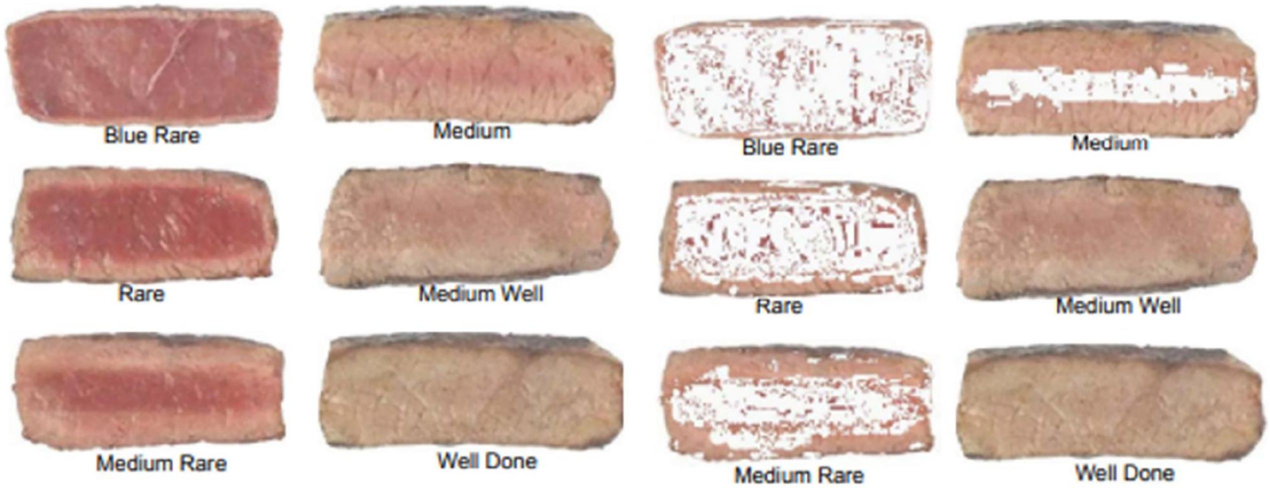

Figure 4. Red and pink highlighting to gauge meat rawness (Tanuwidjaja et al., 2014).

colour wise or cooking and being able to tell when the meat is cooked well as shown in Figure 4.

The device proposed would overlay these items/tasks to allow a natural interaction to the user. Interestingly, specific scenarios around traffic lights were raised, which can be worked out generally by position, however an AR HUD interface that overlays these traffic lights could help people navigate more easily. An example around painted curbs was raised too, where painted red curbs is a no park zone in the US, but this is not possible to see with some colour blindness. These could be overlaid by AR HUD, but they would only propose this to be overlaid on parking scenarios, where a clear intent is made to park either by positioning, destination or some other forms (Tanuwidjaja et al., 2014). AR HUD system must be helpful to the user and aid with colour blindness and not overload the whole setting with overlays everywhere risking a cognitive overload (Ma et al., 2021; Park & Kim, 2013; Wang et al., 2022). It is clear that, AR HUD can help with colour blindness with overlays and perhaps it can help with other conditions that might affect people that drive. Certainly, it could help with a whole host of disabilities up when SAE Level 4 (International, S, SAE J 3016 - Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, 2023) is introduced. After that point it is expected that people won't need to hold a driving license or legally be able to drive. This is a wide topic, one that requires more research and consideration in the future.

5. Design and features

5.1. Obstacle highlight

In another study AR HUD was compared against traditional HUD in recognition of pedestrians along with automated intervention. It showed that AR HUD had way faster recognition of the targeted pedestrian and shown that it was a more acceptable perception of automated intervention. A traditional HUD cannot impose a graphic over or around a pedestrian, so the study actually found that on average a pedestrian warning indicator on a traditional HUD took six steps of their gaze whilst an AR HUD only two, further backing up the faster recognition results (Karatas et al., 2020). Other research with 23 pedestrians found that in every situation the driver noticed the pedestrian with AR sooner (Phan et al., 2016).

The way pedestrians are highlighted is important, as shown in Figure 5. Projected paths (virtual shadow boxes) generally performed significantly better than bounding boxes (Chen et al., 2024). Interestingly, while bounding boxes detected pedestrians more frequently, they detected pedestrians less accurately. This came at the expense of correctly identifying other elements in the environment (Kim & Gabbard, 2022), as shown in Figure 6. For more advanced use cases and in fact to accurately augment pedestrians path, the pedestrians pose, and skeleton patterns extraction should be taken into consideration and using a neural network the intentions of said pedestrian can be determined (Liang et al., 2023).

All of the points raised so far highlight AR HUD being beneficial in recognition of objects, pedestrians and improving safety (Karatas et al., 2020; Ma et al., 2021; Wang et al., 2022). Generally, the results are favourable, with navigation and landmark recognition being praised (Bolton et al., 2015; Phan et al., 2016; Wang et al., 2022; Wu et al., 2009), however it must carefully be implemented to actually improve it over no highlights (Kim & Gabbard, 2022). As raised by various research, inattentional blindness and cognitive load is something that needs to be researched more and taken into consideration during the design and implantation of AR HUD systems (Bauerfeind et al., 2021Phan et al., 2016; Bram-Larbi et al., 2021; Cheng et al., 2023; Ma et al., 2021; Park & Kim, 2013; Schneider, 2021; Teng et al., 2023; Wang et al., 2022). In addition, more research must be conducted around the benefits of AR HUD compared to HUD, since those improvements are not so clear (Feierle et al., 2019; Gabbard et al., 2019; Schneider, 2021). Overall, it seems that these sources agree on the overarching principle; more needs to be understood in how AR HUD interfaces should be designed. One literature gap identified then is more in-depth analysis of different types of conditions and scenarios

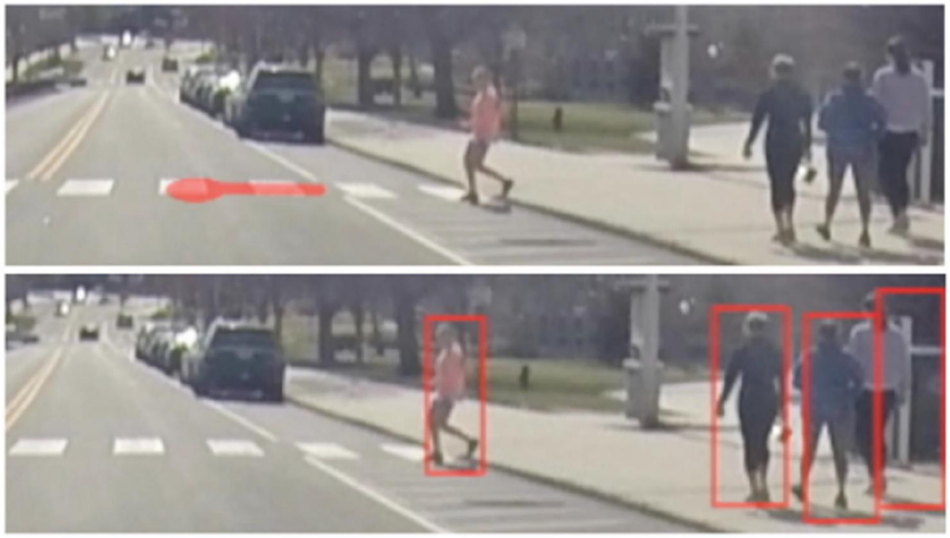

Figure 5. Example of pedestrian highlight and pedestrian projected path (virtual shadow) (Kim & Gabbard, 2022).

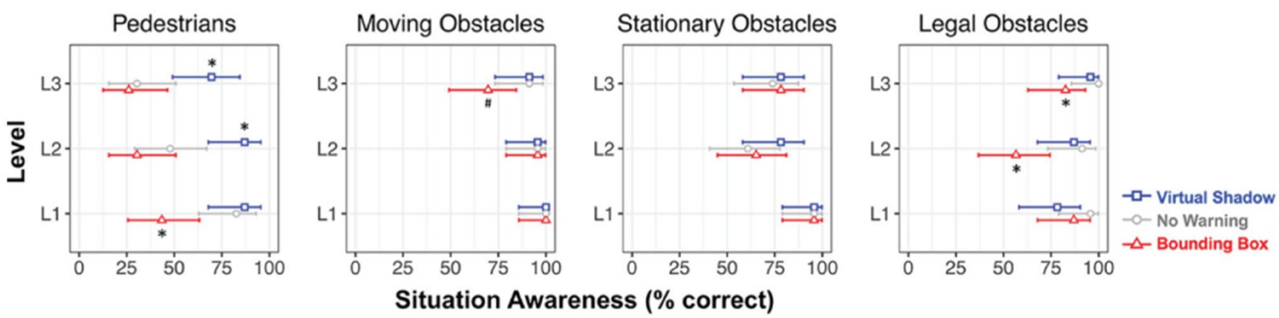

Figure 6. L1 perception, L2 comprehension, and L3 projection of driver awareness of four types of environmental constraints (Kim & Gabbard, 2022).

comparing HUD to AR HUD, and not only that, but several different types of AR HUD. Most of these have focused on object highlight either with pedestrians, road signs or buildings. First of all, these can be tested in a multitude of different ways, but in addition AR HUD can have many other applications, such as additional overlay for difficult to see road marking (Hillen, 2021). Another application as we move towards automated vehicles as an industry is AR HUD applications in automated vehicles.

5.2. Visibility and 3D

An angle to consider around effectiveness is visibility. Specific research by Gabbard et al. around visibility of AR element found that AR HUD graphics are prone to colour blending, which results in intended colour of graphics hard to perceive degrading overall usability. It explored scientific backgrounds and suggested some possible solutions around colour matching (Gabbard et al., 2022). This is another factor to consider and appropriate actions must be taken to ensure the effectiveness of AR HUD doesn't degrade due to issues around colour. Deng et al. highlighted this by discussing the colour gamut differences between HUD's display layers. These can be alleviated somewhat by colour calibration, but in addition ghosting around the border of objects is worth considering, and it is something that should be supressed by some method for example a suggested dynamic distortion calibration which proposes a dynamic process that contains preprocessing, calibration pre distortion and post-processing (Xu et al., 2023). One helpful research in the area highlighted a method that uses object identification and dynamically changes the colour of the content displayed so that it doesn't clash with real world background like a black graphic overlaid with a black car. This or similar method could also help alleviate some of the potential issues (Lee et al., 2021). Image degradation was also discussed, but this is specifically in their research of stereoscopic 3D. As opposed to having 2D images with projections in 3D space, the research focused on 3D projections in 3D space for a more natural feel. Image degradation can happen when the users position shifts when using stereoscopic 3D. A "sweet spot" is highlighted, along with the comments that further work will need to be done to expand this sweet spot and minimise image degradation upon normal movement of the user during driving (Deng et al., 2021). This raises an important distinction between 3D and 2D projections in the application of AR HUD. The information collected shows that stereoscopic depth cues are more useful than stereoscopic AR HUD applications and in fact monoscopic cues are effective between 1.5 to 30 m, which matches up with where AR HUD would start projecting – beyond the cars bonnet. Therefore, it is argued that for automotive

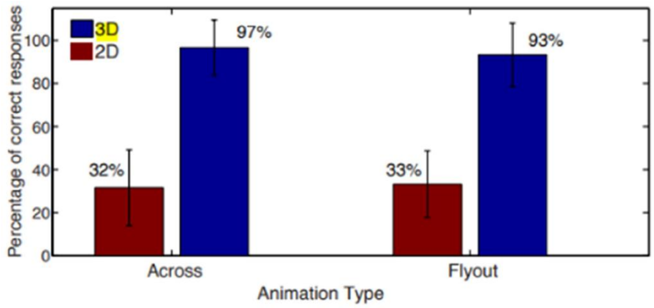

Figure 7. 2d vs 3d comparison in the ability to judge distance (Bark et al., 2014).

applications monoscopic display is suitable (Gabbard et al., 2014). In contrast however Bark et al. research indicated that 3D system was much easier to determine what stanchion a virtual airplane was flying close to in an experiment. In fact, 3d seemed to have a significant effect in judging distance, as illustrated in Figure 7 (Bark et al., 2014).

As mentioned by Deng et al. when we aim to implement a stereoscopic 3D AR HUD, we must look out for the proposed sweet spot in which the 3D object, in fact, appears 3D (Deng et al., 2021). Not meeting this sweet spot could cause the image to shift, conveying no visibility or incorrect depth perception, which could have severe outcomes, when important driving information or task is displayed. Stereoscopic 3D also requires more performance, as depending on the solution two images must be rendered.

5.3. Size and controls

The interaction and the size of the projected HUD image is crucial too. Charissis et al. (2021) in their research found that a medium sized HUD will perform better compared to a more traditional smaller HUD, utilising more of the windshield real estate. In general, all features of the AR HUD that this literature view has explored would suffer from a smaller size. For example, landmark boxes or arrows that project an upcoming turn would not be able to be displayed further away to the side of the scene if the size of the AR HUD does not support it. Or an augmented lane keep assist could only project lane markings on a straight road, since as soon as the lane turns, the size of the AR HUD cannot accommodate until the vehicle start turning towards the bend and lining up the front of the car with the turn. To better illustrate the issue, Figure 8 has been overlaid with a green square showing a larger size AR HUD that accommodates turns and lanes as well as upcoming landmarks and a red box showing the issue with a smaller AR HUD. The red box illustrates what happens when an AR HUD is used that is much too small to cater for upcoming turns for instance. The scene cannot be augmented to a good degree in that example arrow and lane cannot be shown properly due to the small AR HUD size. Another issue would be traffic and pedestrians. All the positive about augmenting turns, cars, pedestrians and obstacles reduce if these can only be augmented at a narrow field of view straight ahead of the car. A human factors study explored sizing too, where they defined specific size and angle requirements for text to be legible (Wan & Tsimhoni, 2021). The larger the AR HUD, the better these sizes can be adhered to. It is suggested that even warning symbols need to be a certain size to be effective (Li et al., 2023), therefore the AR HUD size should make this possible. One research found that the field of view (not the overall size) did not have a significant effect on situational awareness (Pečečnik et al., 2023). Certainly, more research is required around ideal sizes of AR HUD interfaces and the effects of different sizes.

Visibility is one aspect, how users interact with such system should be kept to a minimum with no “complicated menus and buttons positioned in the HDD space or awkward to reach positions” (Charissis et al., 2021). For interactions gestures recognition has been proposed as one of the solutions, which might have some benefits without potentially affecting visual and cognitive attention. Based on the implementation, gesture controls can have benefits around not needing the user to focus on a specific point of a touchscreen or button and can do it without that specific focus on that point. One negative is learning – as there would be a greater learning required in order to learn specific gestures, how to do them correctly, what each one means, but also recalling these gestures (Charissis et al., 2021). Based on these results show that users can quickly learn up to six gestures, however these gestures should correspond to movements similar to interpersonal communications (Graichen et al., 2022). There is more of a disconnect between a gesture achieving an action and a labelled button for instance, therefore more research and trials must be completed to review the effectiveness of gesture based controls.

6. Unaddressed challenges and future trends

One of the literature gaps is around disabilities, for instance if the AR HUD has 3D projection and it conveys important information, what about users that do not have the ability to view 3D content or have reduced/no depth perception? How would a 3D AR HUD cater? This is simply one aspect that must be considered when looking at 3D AR HUD or even AR HUD as a whole.

The visibility of HUD must be researched more, for example with sunglasses, which was raised as a potential issue within JLR previously. Another topic would be the overall design of the interior of the vehicle, that must change to accommodate an interior space with no driver display. Along with this change, it is likely that controls might need researching. The reason being users might expect different controls when interacting with large AR HUD only interfaces. But this expectation must be researched. What feels right in an AR HUD only interface? For example, the gesture-type controls suggested (Charissis et al., 2021) and their impact. Nielsen can give us some basic heuristics that can be used like the visibility of system status, error prevention recognition rather than recall and so on (Nielsen, 2023), but how do these apply to gesture based control? Research around gestures provides some clear

Figure 8. AR HUD showing lane and upcoming turn (All You Need To Know About AR Heads Up Displays, 2023) with overlay added illustrating possible AR HUD sizes.

points such as reducing gesture constraint rates to ensure more success in completing gestures (Gable et al., 2014), but still gestures is simply one possibility and detailed research should take place to see how people prefer to interact with AR HUD only interface. There seems to be some gap around psychological factors, as one research suggested that AR HUD could help with anti-personal anxiety (Hwang et al., 2016). This could be looked at in the context of this AR HUD only interface. More research into passenger AR HUD display is needed. A paper exploring different concepts found that passengers would benefit from AR HUD to displaying navigation information to aid the driver or understand path a taxi takes have demonstrated a high demand (Zhu et al., 2024). As well as passengers it has been noted that pedestrians could benefit from AR interfaces to increase the safety for everyone on the roads (Tabone et al., 2023).

The main literature gaps I have identified and will focus on are identified below. It is worth noting that two main gaps have been identified, however they are closely related. Size requirements must be understood in order to use AR HUD as a main display.

6.1. AR HUD as main display

Based on the available research this literature review has explored the evolution of car displays, HUD and AR HUD. It compared HUD against AR HUD and explored the effectiveness of it. All of the found research has one aspect in common: a primary display. This main display would typically show the speed, and any other main function related to driving, be it on the digital driver display or a centre display. This means that HUD and AR HUD research usually focuses on features that are an expansion of the main display, which other than the speed, could usually provide navigation, traffic information or mobile phone interaction such as calls (Maroto et al., 2018). The literature gap identified here is around main display and what if we take an AR HUD and use it as a main display instead? Is it possible? Von Sawitzky et al. in their research found that the only information that they found useful to be shown is world-fixed information is information that relates to a specific turn or landmark for instance (Von Sawitzky et al., 2019), so further research on adding all the extra content from the digital driver display would be useful. Figure 9 effectively highlights this gap in the literature: what if we move the "secondary task" into the field of the "primary task," entirely eliminating the need for a separate secondary task display?

One very important aspect to make here is the size of the AR HUD interface, as illustrated in Figure 8, the wrong size interface simply might not have the capability to highlight pedestrians, obstacles or effectively handle a handover which were some of the major positives identified. There does not seem to be clear research on the effects of an AR HUD interface that is too small. What is too small and what are the effects? What about one that is too large? Size needs to be researched to a greater extent to ensure that AR HUD can effectively be used as a main display.

Currently in JLR vehicles we can display "vehicle speed, navigation, cruise control notifications and Traffic Sign Recognition data" (New Range Rover Velar, 2024) and additionally based on my work with HUD and the driver information systems team within JLR I can confirm it can display speedo, gear, media, phone data (like calls), critical warning (full size telltale) 4x4i (display offroad widget like angle and differential settings).

The first thing that is worth identifying is the type of features that would in addition, need to be part of this exclusive AR HUD solution other than the list above. Table 4 contains some of the main features typically found in a JLR vehicle and considerations around it. It is worth noting that most of these might have research applicable to it, for example telltale legislations might have research and so is the visibility of HUDs and its effectiveness. However, the two together might require more research.

In addition to some research required on individual items, there must be more research in the overall impact of AR HUD only solution. As shown in Table 4, there are tons of content that must be added to AR HUD that are not found in today's vehicles. Therefore, it can be argued that the size of the AR HUD area would need to be a lot larger to accommodate. Consideration around the required AR

Figure 9. Primary, secondary and tertiary tasks in vehicles cabin (Tonnis et al., 2006).

Table 4. List of features that would move to AR-HUD only solution.

Content	Considerations
Telltales	The issue around telltales is that they are usually ASIL rated and must match strict legislation for legibility, visibility, location amongst others. Gabbard et. al highlighted the issue around colour blending (Gabbard et al., 2022) and in general from a personal experience I had problem with seeing specific colours in HUD of JLR vehicles, which has been discussed along with the Human Factors UX team. This is something that would need more research directly in relation to legally required telltales showing on HUD.
Lights	Part of telltales
Indicators	Part of telltales
Temperature	Text is already in some HUDS with media and phone information, outside and inside temperature could be displayed fine. However, when looking at temperatures, cabin temperature should also be considered. In that case, 2 zone, or even 4 zone temperature might need to be displayed within a HUD-only solution, risking too much information being displayed to the driver and going back to cognitive overload considerations (Ma et al., 2021; Park & Kim, 2013; Wang et al., 2022).
State of charge	State of charge and Range can go together. Both of these should have no
Range	additional consideration as they are simple information to present on HUD.
ADAS confidence view	This might need a lot of research. Out of everything listed, this is perhaps the most dynamic with constantly moving and changing scene. Potentially this would need to be up at all times too, during automated driving. Elements to research around visibility and effectiveness.
Eco coaching	Related to telltales.
Notifications, warnings and errors	Currently in JLR vehicles we only have large red triangle in HUD during critical warnings. Additionally, to that, the HUD-only solution would require all additional messages to be included, so the impact of more text to read and smaller graphics must be researched. Additionally, translations of text must be considered in terms of visibility and size. For instance, Arabic text might not display as well on HUD and certainly takes up more vertical space.
ADAS modes	Simple toggle between ADAS modes like low and high. Related to notification, warnings and errors.
Driving modes	Text and graphic. Toggle between different driving modes like Comfort and Eco.
Volume	No consideration.
Door open warning	Perhaps larger graphics for these but would be related the research around warnings with graphics.
Belt reminder	Related to telltales.

HUD size to fit all content might have an effect on the whole vehicle experience and safety. Research required around an AR HUD that takes up a larger part of the windshield in terms of experience, distraction and effectiveness. Cognitive overload must be considered as well as overall safety and visibility. If the windshield has so much content on it, what if it blocks critical real life driving information?

Overall, the current research state focuses largely on digital driver display compared to HUD or HUD compared to AR HUD. The effectiveness and safety of AR HUD is also researched. Some research findings contradict each other, but generally AR HUD is positive from the research gathered. However, all the found research results might or might not apply when we look at an AR HUD only solution, therefore AR HUD in the context of it being the only display must be researched further.

Augmented Reality Heads-Up Display is a large topic as illustrated by this literature review. It covers various elements from the comparison of traditional fixed HUD vs AR HUD where AR HUD showed to be better (Bolton et al., 2015), however in contrast found that the system must be implemented in careful way, since it can have the opposite effect with a badly designed AR HUD system that increases workload (Gabbard et al., 2019). Most research agreed that AR HUD applications must use landmark elements in order to be effective (Bark et al., 2014; Wang et al., 2022; Wu et al., 2009), and in fact the system application must be extremely careful to ensure it does not cause inattentional blindness by taking the focus away from elements such as pedestrians that should be augmented (Karatas et al., 2020; Wang et al., 2022). However, highlighting the right parts of the scene is one aspect, because a good interface that follows solid human technology interaction principles can boost safety and cognitive load (Ma et al., 2021), but a bad design would have the opposite effect (Ma et al., 2021; Park & Kim, 2013). Further to this, specific information must be displayed to increase trust (Abdi et al., 2015; Von Sawitzky et al., 2019). AR HUD came out on top of for handovers between automated and manual driving too (Langlois & Soualmi, 2016). The literature review highlighted other considerations such as psychological factors (Hwang et al., 2016), colour blending (Gabbard et al., 2022), 2D vs 3D display considerations (Gabbard et al., 2014), control methods such as gestures (Charissis et al., 2021) to name a few.

7. Conclusion

In conclusion, the results show that AR HUD can be an excellent system in today's vehicles, boosting user experience by reducing cognitive load, increasing trust, and improving safety. However, there is still much research required to fully realise its potential. At its best, AR HUD can reduce cognitive load, increase user trust, and enhance general safety. Conversely, if not implemented correctly, AR HUD can increase cognitive load, worsen navigation, and decrease safety.

It is evident that more needs to be done to ensure the successful application of AR HUD. Some basic applications are already available in select vehicles, with more expected to come. Therefore, it is important for the industry to conduct additional research and studies. As more automated vehicles emerge, the key benefits of AR HUD, such as facilitating actions like handovers, have been highlighted.

The ability of AR HUD to provide more inclusive design options, such as assisting with colour blindness and incorporating alternative controls like gestures, could be highly beneficial for the industry. The literature gap identified is using AR HUD as the only display within the vehicle. This concept, while challenging, has the potential to significantly enhance user experience, much like the transition from traditional dials to fully customisable digital driver displays.

Future research should focus on optimising the design and implementation of AR HUD to maximise its benefits while minimising potential drawbacks. This includes investigating the impacts on cognitive load, safety, and navigation in various driving scenarios, especially in automated vehicles. Additionally, exploring user interactions and preferences in different demographic groups, including those with disabilities, will be crucial for developing more inclusive AR HUD systems.

Overall, the integration of AR HUD in vehicles promises to transform the driving experience, making it safer, more intuitive, and more enjoyable. Continued research and development in this area will be essential to unlock the full potential of AR HUD technology.

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